

Chandigarh University

UNIVERSITY INSTITUTE OF COMPUTING

MINI PROJECT / CASE STUDY

E-COMMERCE SALES DATA ANALYSIS

SUBMITTED BY.

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1. Problem Statement

E-commerce platforms generate large amounts of transaction data. This case study investigates which products and categories contribute most to sales, how revenue changes over time and across regions, and what relationships exist among quantity, price, discount, customer age, rating and revenue.

2. Dataset Description

The raw dataset contains 602 transaction records. After data cleaning, 600 unique records were used for analysis. The dataset contains order details, product information, pricing, discount, revenue, payment method, region, customer age and rating.

Variable	Data Type	Description
Order_ID	Object	Unique order identifier
Order_Date	DateTime	Order date
Category	Object	Product category
Product	Object	Product name
Quantity	Integer	Units purchased
Unit_Price	Float	Price per unit
Discount	Float	Discount fraction
Revenue	Float	Revenue per order
Payment_Method	Object	Payment method
Region	Object	Customer region
Customer_Age	Integer	Customer age
Rating	Float	Customer rating

3. Data Cleaning Using Pandas

The raw data contained 3 missing cells and 2 duplicate records. Pandas was used to prepare the dataset for reliable analysis.

- Missing values: missing ratings were filled with the median rating.
- Duplicate records: duplicate rows were removed.
- Incorrect data types: date and numerical fields were converted to suitable types.
- Inconsistent values: region and payment-method labels were standardized.

4. Data Analysis Using NumPy & Pandas

Measure	Result
Total orders	600
Total units sold	1232
Total revenue	₹1,764,544.40
Mean revenue/order	₹2,940.91
Median revenue/order	₹2,067.22
Standard deviation	₹2,484.55
Average rating	4.05/5

Category-wise revenue:

- Electronics: ₹574,938.35; 167 orders; average order revenue ₹3,442.74.
- Home & Kitchen: ₹419,235.00; 111 orders; average order revenue ₹3,776.89.
- Clothing: ₹401,046.70; 136 orders; average order revenue ₹2,948.87.
- Sports: ₹251,172.45; 96 orders; average order revenue ₹2,616.38.
- Beauty: ₹118,151.90; 90 orders; average order revenue ₹1,312.80.

Highest-revenue category: Electronics. Highest-revenue product: Smart Watch. Highest-revenue month: 2025-04. Highest-revenue region: West.

Correlation between Quantity and Revenue: 0.64.

5. Data Visualization

Matplotlib was used for line, bar, histogram, pie and scatter visualizations. Seaborn was used for box plot, violin plot and correlation heatmap visualizations.

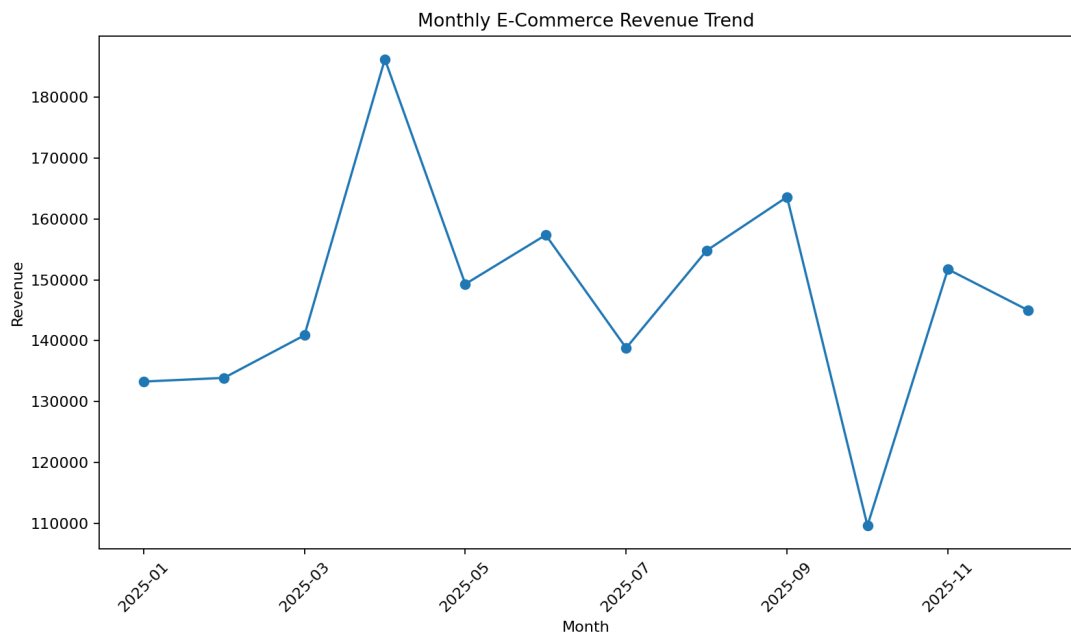


Figure 1: Monthly Revenue Trend

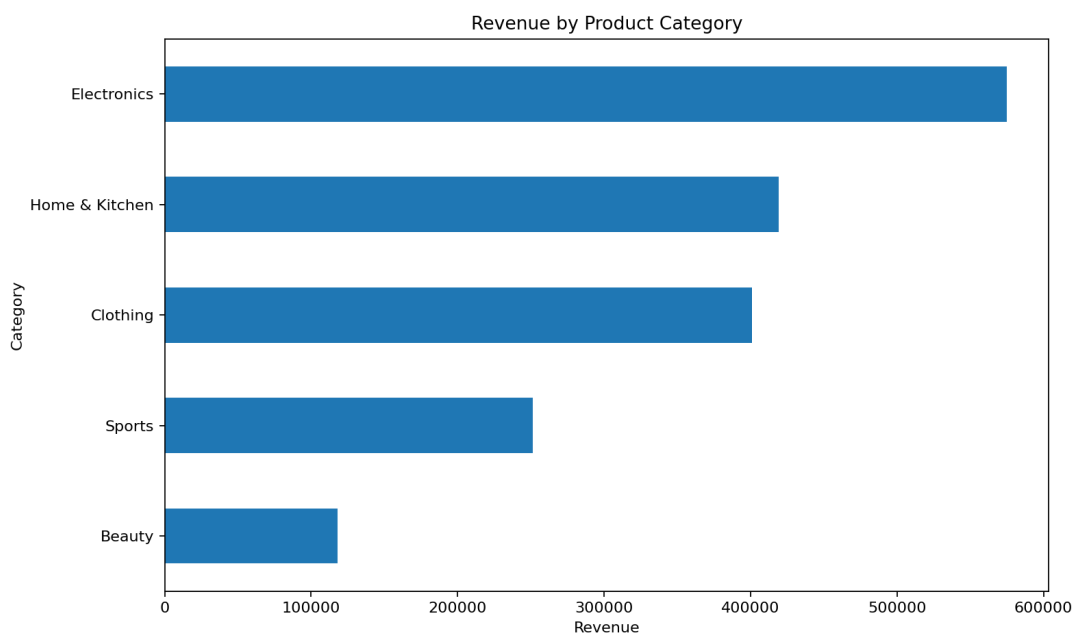


Figure 2: Revenue by Product Category

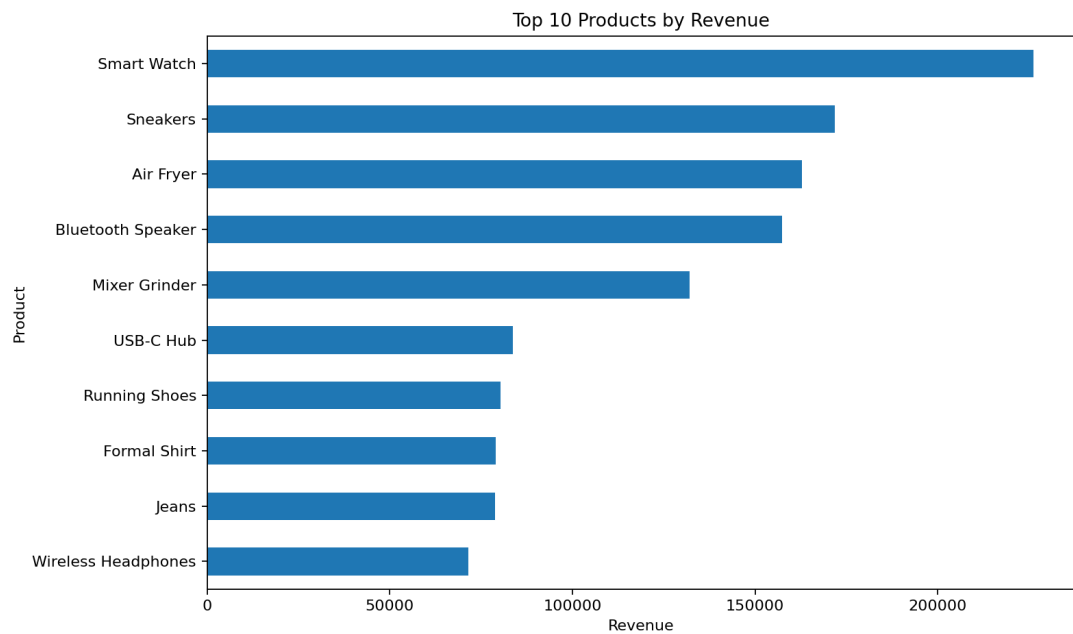


Figure 3: Top 10 Products by Revenue

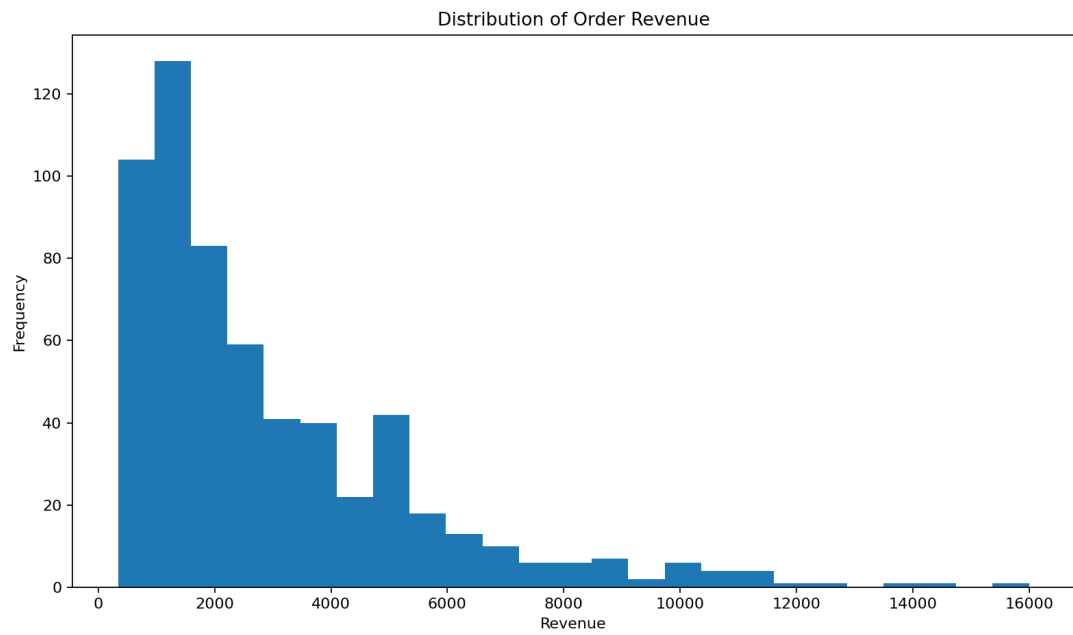


Figure 4: Distribution of Order Revenue

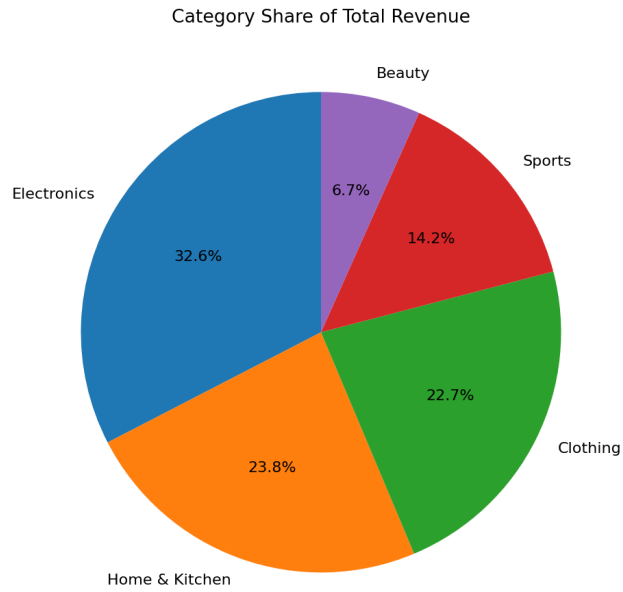


Figure 5: Category Share of Total Revenue

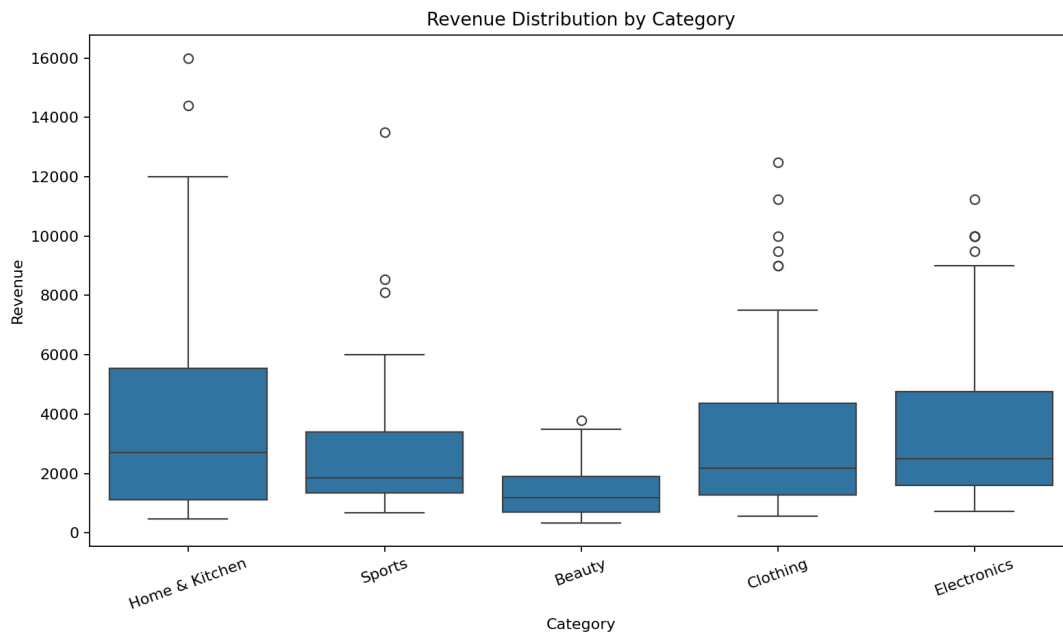


Figure 6: Revenue Distribution by Category

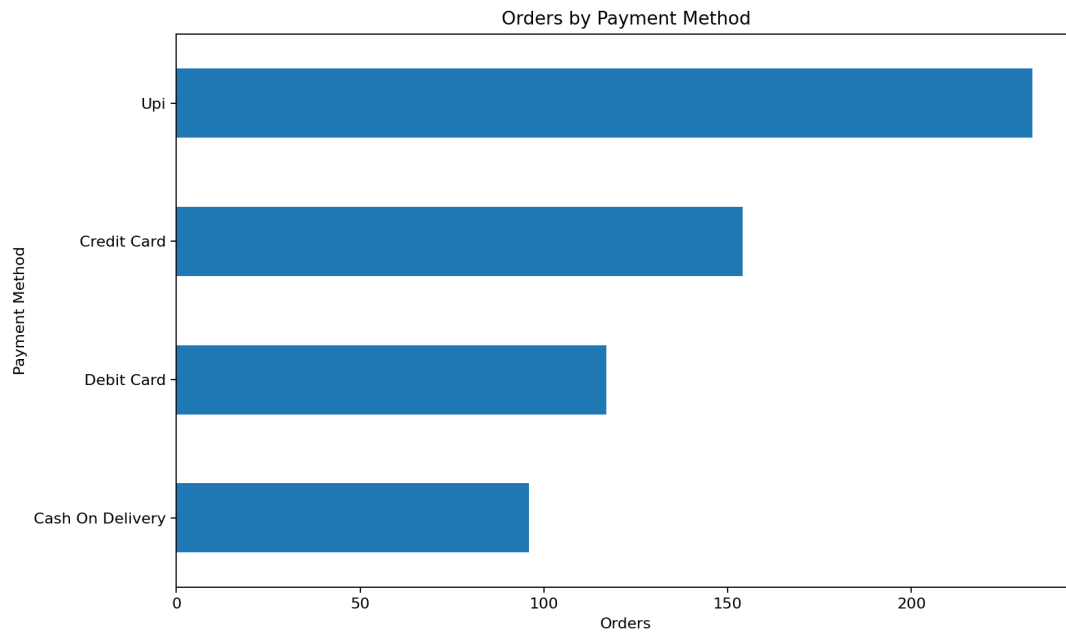


Figure 7: Orders by Payment Method

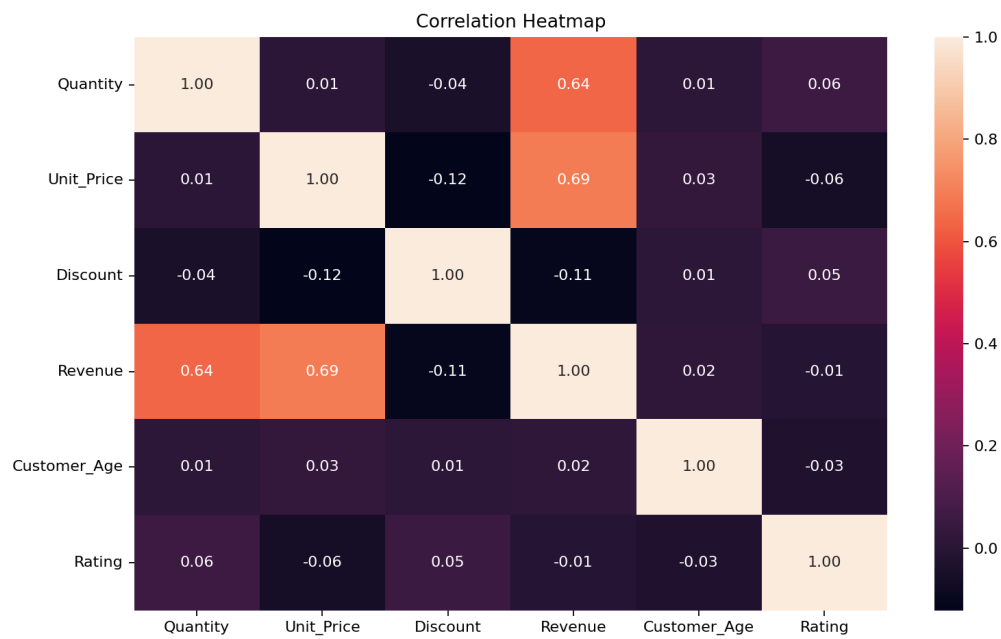


Figure 8: Correlation Heatmap

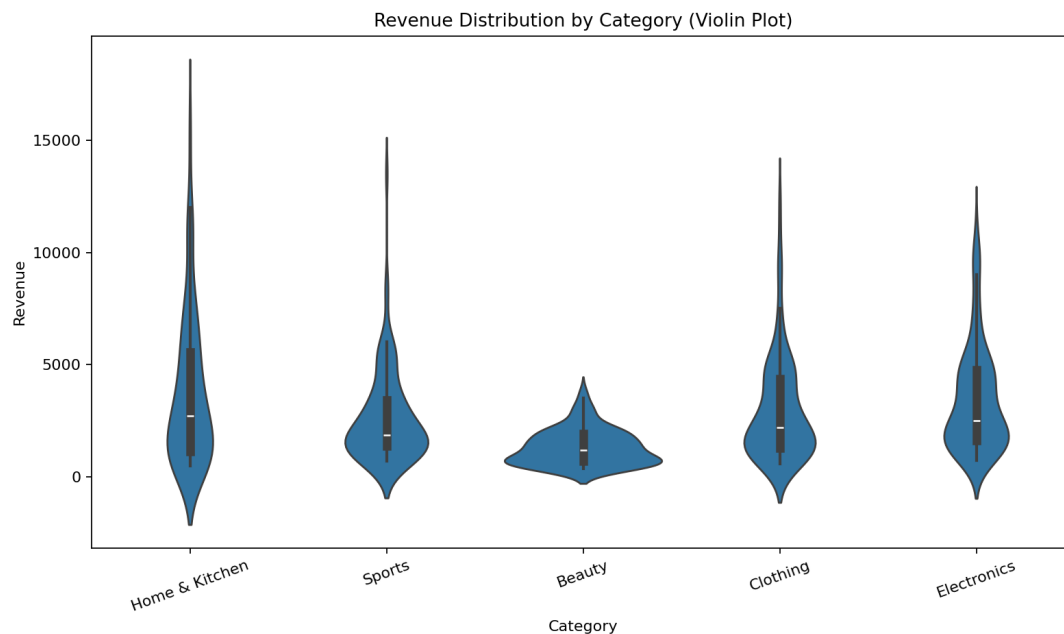


Figure 9: Revenue Distribution by Category

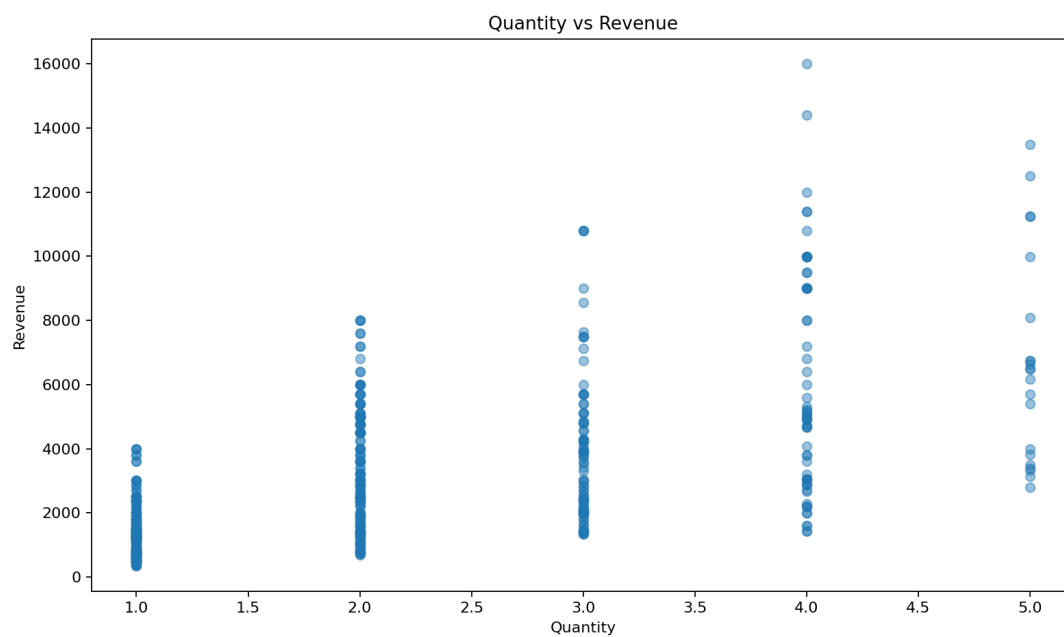


Figure 10: Quantity vs Revenue

6. Findings

1. Electronics generated the highest category revenue of ₹574,938.35, approximately 32.6% of total revenue.

2. Smart Watch was the highest-revenue product.
3. 2025-04 recorded the highest monthly revenue.
4. West generated the highest regional revenue.
5. Mean revenue per order was ₹2,940.91, compared with a median of ₹2,067.22, showing variation in order values.
6. The average customer rating was 4.05/5.
7. Quantity and revenue had a positive correlation of 0.64.
8. Upi was the most frequently used payment method.

7. Conclusion

The analysis demonstrates how Pandas, NumPy, Matplotlib and Seaborn can be used to convert transaction data into useful business insights. Electronics was the strongest category, Smart Watch was the strongest product by revenue, 2025-04 was the strongest month, and West led regional revenue. The combined use of cleaning, descriptive statistics, grouping, correlation and visualization provides a clear understanding of e-commerce sales performance.

Tools and Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn